PYTHON ACTIVITY-BASED LEARNING

Case Study 06: Tuple-Based Transaction Audit

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Problem Statement

Audit transaction records stored as tuples and identify duplicate IDs, invalid amounts and transaction types.

Objectives

• Store transaction details in tuples.

• Validate amount and transaction type.

• Find duplicate transaction IDs.

Assumptions

• Each tuple contains ID, amount and type.

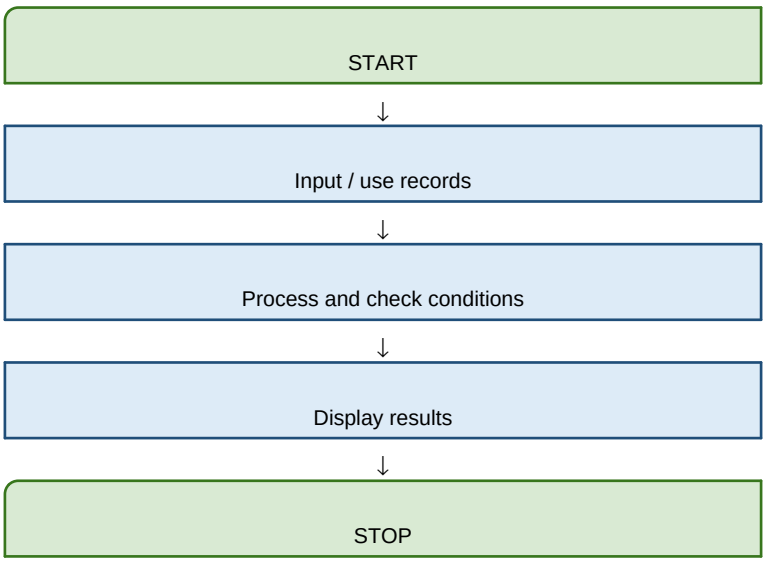
• Valid types are Deposit and Withdrawal.

• Amounts must be greater than zero.

Algorithm

1. Create a list of transaction tuples.
2. For each tuple, unpack ID, amount and type.
3. Check whether ID is repeated.
4. Check whether amount is positive and type is valid.
5. Print the validation result for each record.

Flowchart



Pseudocode

```
START
STORE transaction tuples
SET seen IDs to empty list
FOR each tuple
    UNPACK ID, amount, type
    CHECK duplicate ID
    CHECK amount > 0
    CHECK type is Deposit or Withdrawal
    DISPLAY errors or valid record
STOP
```

Python Source Code

```
transactions = [
    ("A01", 1500, "Deposit"),
    ("A02", 500, "Withdrawal"),
    ("A01", 700, "Deposit"),
    ("A03", -200, "Deposit"),
    ("A04", 300, "Transfer")
]

seen = []

for tid, amount, kind in transactions:
    errors = []

    if tid in seen:
        errors.append("Duplicate ID")
    else:
        seen.append(tid)

    if amount <= 0:
        errors.append("Invalid amount")

    if kind not in ("Deposit", "Withdrawal"):
        errors.append("Invalid type")

    if errors:
        print(tid, "-", ", ".join(errors))
    else:
        print(tid, "- Valid")
```

Test Cases

Test input / condition	Expected result
A01, 1500, Deposit	Valid
A01 repeated	Duplicate ID
A03 amount -200	Invalid amount
A04 type Transfer	Invalid type

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
A01 - Valid
A02 - Valid
A01 - Duplicate ID
A03 - Invalid amount
A04 - Invalid type
```

Conclusion

The program demonstrates tuple unpacking and conditional validation for transaction records.